Testing Portfolio Exposure from Text with Matched Holdings

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Abstract

When every product in a portfolio holds the same percentage of a target asset, even incorrect position weights can yield the correct exposure. We construct 24 pairs of portfolio questions with identical investment amounts and the same correct total. Each pair contrasts products with identical target allocations against products with different allocations. A greedy Qwen3-8B configuration answered all 24 identical-allocation questions correctly and 4 of 24 varied-allocation questions correctly. A reasoning-enabled configuration of the same checkpoint answered all 48 correctly. Of 26 ledgers requested on published fund tables, 18 contained accurate fields but an incorrect total under the original configuration. Recomputing the reported fields recovered all 18 totals. Across the fixed 100 prompts, the configurations scored 35 and 98. Both remaining failures exhausted the output budget on one split account. Matched holdings test sensitivity to investment amounts, while ledger scoring distinguishes errors in the reported inputs from errors in their aggregation.

1 What does a correct total tell us?

An investor puts GBP 2,500, 1,500, 4,000 and 2,000 into four products. Suppose each product holds 55% of Asset A. A system that reads every position, one that drops the largest, and one that weights the four products equally can all report 55%. The answer alone supplies no evidence that the four investment amounts were used correctly.

Table 1 gives the first family in our evaluation. Keeping the amounts and the true total fixed, we change the four product percentages. Equal product weighting now yields 46.75%, more than eight percentage points below the truth. On this exact text, the original model output was 44.44%. That number differs from both the correct answer and the equal-weight shortcut. The example separates a property of the question from a hypothesis about the model's internal calculation.

	Product 1	Product 2	Product 3	Product 4	True total
Position value (GBP)	2,500	1,500	4,000	2,000	10,000
Same target allocation (%)	55	55	55	55	55
Varied target allocations (%)	67	27	78	15	55

Table 1. A matched family. Qwen's original direct answers were 55% and 44.44%, respectively. The recheck returned 55% for each.

Two observations motivate the evaluation. A test of weighting needs cases in which the weights affect the answer. A test of calculation needs the reported inputs as well as the final number. We use matched holdings for the first purpose and ask for a per-lot ledger for the second.

2 Which weighting errors can the questions reveal?

If v_i pounds are invested in product i and a_i is its target allocation, the account exposure is

$$\theta = \sum_i q_i a_i, \qquad q_i = v_i / \sum_j v_j. \tag{1}$$

The output is a percentage when a_i is expressed in percentage points and a fraction when a_i is expressed as a fraction. The contrast experiment in this section uses fractions. The matched experiments use percentages.

Why an error can disappear. Let p be a different set of nonnegative product weights that also sum to one. For example, omitting a lot and renormalizing the remaining amounts changes q to p. The error on composition a is $(p-q)^\top a$. If $a = c\mathbf{1}$, this equals $c(\sum_i p_i - \sum_i q_i) = 0$. The cancellation holds for every normalized p. Even a varied composition can miss an error if its positive and negative contributions cancel.

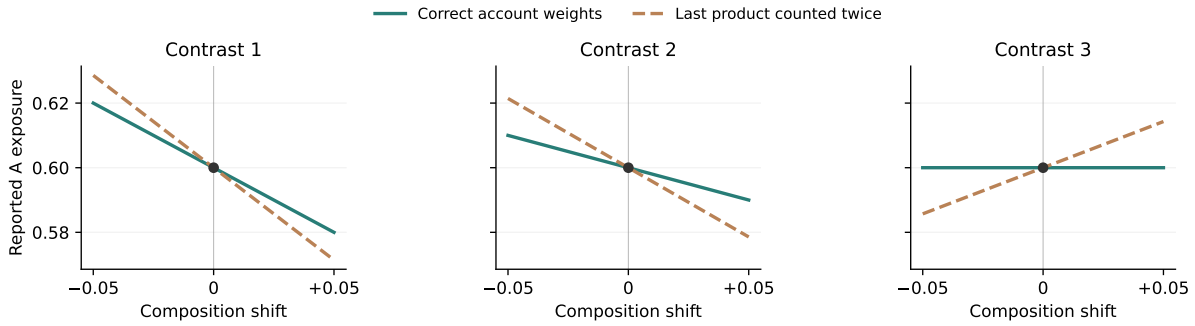


Figure 1: Two specified weighting rules applied to the same account. Starting from 60% target holdings in every product, three different composition changes separate correct weighting from counting the last product twice. The lines are calculated from these rules.

What an informative contrast adds. Figure 1 uses account weights $(.1, .2, .3, .4)$. In the first contrast, add five percentage points to the first two product allocations and subtract five from the last two. The correct exposure falls from 60% to 58%. Counting the last product twice gives a new exposure of about 57.14%. The two rules agree at the centre but respond differently to the same edit.

For a fixed rule $f(a) = p^\top a$, the centred response to a change along c is $[f(a+\delta c) - f(a-\delta c)]/(2\delta) = c^\top p$. Collect the contrast rows in C. Two normalized weight vectors can agree on every contrast only when their difference e satisfies $Ce = 0$ and $\mathbf{1}^\top e = 0$. Thus normalization and contrasts identify every fixed weight vector exactly when the stacked rows have full column rank. If rank falls short, a sufficiently small null-space displacement from an interior weight vector gives another admissible answer. Four products require at least three independent contrasts alongside normalization. Applying this reasoning to a language model requires the same effective weights across all edited prompts.

How large could the hidden error be? Write $e = p - q$ and let each product allocation range over $[\ell, u]$. Positive and negative entries of e have the same total magnitude because $\sum_i e_i = 0$. Setting $a_i = u$ on positive entries and $a_i = \ell$ on negative ones attains the maximum absolute exposure error, $(u-\ell)\|p-q\|_1/2$. This gives the size of an error a homogeneous question can hide under the fixed-rule assumption. In a deterministic check on 3,744 injected weighting distortions, one tested contrast detected 3,648 while the three-contrast basis detected all 3,744. These counts describe specified arithmetic rules, separately from the model responses below.

3 Testing the fixed-weight assumption on recorded answers

An earlier diagnostic study used 16 families and 32 cases per configuration on SmolLM3-3B and Qwen3-8B [6, 9]. It compared a contrast check with a consistency check on the same base answers. The contrast check requested the base answer, six answers to three symmetric composition edits, and an all-B anchor. The comparator used the base, six lot permutations, and a split-lot description. Each therefore cost eight requests. For valid fractional answers, the contrast check allowed slope discrepancies of 0.02, midpoint discrepancies of 0.005, and an anchor error of 0.005. The comparator required valid answers within a range of 0.005.

Configuration	Correct base answers	Consistency accepts	Contrast accepts
SmolLM3-3B	10/32	2/32	0/32
Qwen3-8B	13/32	1/32	0/32

Table 2: Earlier diagnostic study. All three answers retained by the consistency check were correct. The contrast check rejected every base answer, including all 10 and 13 correct answers in the respective runs.

A recorded counterexample. In a post-hoc check, one Qwen case answered its base question correctly and gave target fractions .246, .346 and .346 on three positive contrasts. Each contrast starts from four identical fractions .296 and shifts by .05. The respective sign patterns are $(+,+,-,-)$, $(+,-,+,-)$ and $(+,-,-,+)$. Calling the resulting input vectors x_1, x_2, x_3, the largest coordinate of $-x_1 + x_2 + x_3$ is .346. Every normalized nonnegative p therefore satisfies $p^\top(-x_1 + x_2 + x_3) \leq .346$. The corresponding combination of reported answers is .446. If a fixed rule fits each report within ϵ, then $3\epsilon \geq .100$. At least one report misses that rule by at least 3.33 percentage points. This case was the sole extra Qwen exclusion from the full feasibility check beyond the single-query, two-query and midpoint bounds. Table 2 shows why the rule had no practical coverage as an acceptance filter.

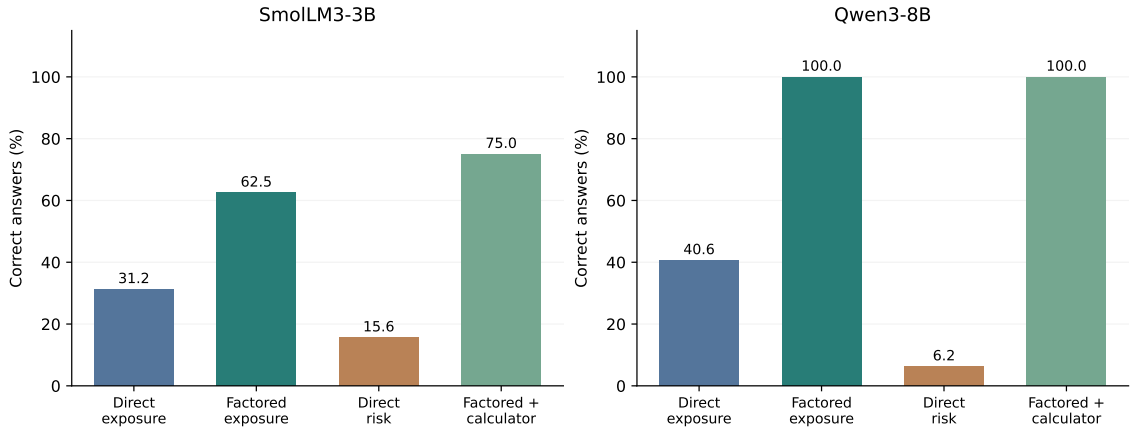


Figure 2: Earlier study, 32 cases per configuration. The exposure columns compare direct answers with extraction of per-lot source fields. The risk columns compare a direct risk answer with calculation from per-lot fields under supplied return scenarios. Exposure tolerance was 0.5 percentage points and risk tolerance was 5% relative error.

Figure 2 gives a reason to request explicit fields. For Qwen, direct exposure was correct on 13 cases and per-lot extraction gave correct exposure on all 32. This motivated a separate field score in the subsequent experiments. The 32-case study has its own prompts and denominators.

4 Matched accounts and published fund tables

Controlled accounts. Each of 24 designed families contains four unequal investment amounts. One question uses identical product allocations. Its pair uses four distinct allocations with the same correct total. In every heterogeneous case, an unweighted product mean misses the true answer by at least five percentage points. All 48 prompts ask directly for the total and present each product position as a single funded lot.

Published holdings. We use 88 holding-name and percentage rows from eight funds in four Vanguard and BlackRock documents dated September 2026 [2, 11]. We form all ten within-Vanguard fund pairs and all three within-BlackRock pairs. Each constructed GBP 10,000 account has two presentations, one with a position split into separately funded lots. Each presentation is queried twice, once for a total alone and once for a per-lot ledger and total. This gives $13 \times 2 \times 2 = 52$ prompts. Tables were transcribed and standardized before inference. In each prompt the target is a named underlying fund, and absence of that name from the supplied table counts as zero. Table 3 gives the inputs for each named target.

Paired configuration comparison. We reused the saved original responses to all 100 inputs and collected one new response per input. Both configurations use the same pinned Qwen3-8B Q4_K_M checkpoint [9], system and user text, and JSON-object constraint. The original used greedy decoding, disabled thinking and a 512-token output cap. The second enabled thinking and used temperature 0.6, top-p 0.95, top-k 20, min-p 0 and up to 4,096 output tokens. The seed, context length and batching also changed. The recheck protocol was frozen before its evaluation calls, after we had examined the original answers. Four engineering calls on separate prompts are excluded. Each condition contributes one response per prompt.

Scoring. The primary score uses the requested JSON and the final answer field. A total must lie within 0.5 percentage points of the accounting value. A correct ledger must contain exactly the expected lot IDs, each amount within GBP 0.005 and each allocation within 0.00005 percentage points. An external calculator applies Equation 1 to the amounts and percentages *reported by the model*. Responses cut off at the token limit count as failures. The 100 prompts reuse 24 families and 13 accounts. All rates below describe this fixed set.

Parent fund	PDF page	Holding rows	Target (%)
LifeStrategy Global 20% Equity	6	12	0.00*
LifeStrategy Global 40% Equity	10	16	6.80
LifeStrategy Global 60% Equity	14	14	14.80
LifeStrategy Global 80% Equity	18	8	28.30
LifeStrategy Global 100% Equity	22	8	33.60
MyMap 3 (Class S Inc)	3	10	13.08
MyMap 4 (Class S Inc)	3	10	17.89
MyMap 5 (Class D Inc)	3	10	17.88

Table 3: Source data for the constructed accounts. The target is Vanguard U.S. Equity Index Fund in the LifeStrategy report and ISHRS US EQ IDX FD (UK) X ACC in the MyMap factsheets. *The named target is absent from the supplied Global 20% table and contributes zero under the task definition.

5 What the two original configurations showed

The original controlled experiment evaluated 24 families on Qwen3-8B and Phi-4 14B. Each family supplied one presentation with four funded positions and one with six lots after splitting a position. For each type of composition, the resulting 48 direct prompts comprise 24 related presentation pairs. Both configurations used greedy decoding and a 512-token output cap. Their architecture and quantization differed. Qwen used Q4_K_M and Phi used Q3_K_S [1, 7, 9].

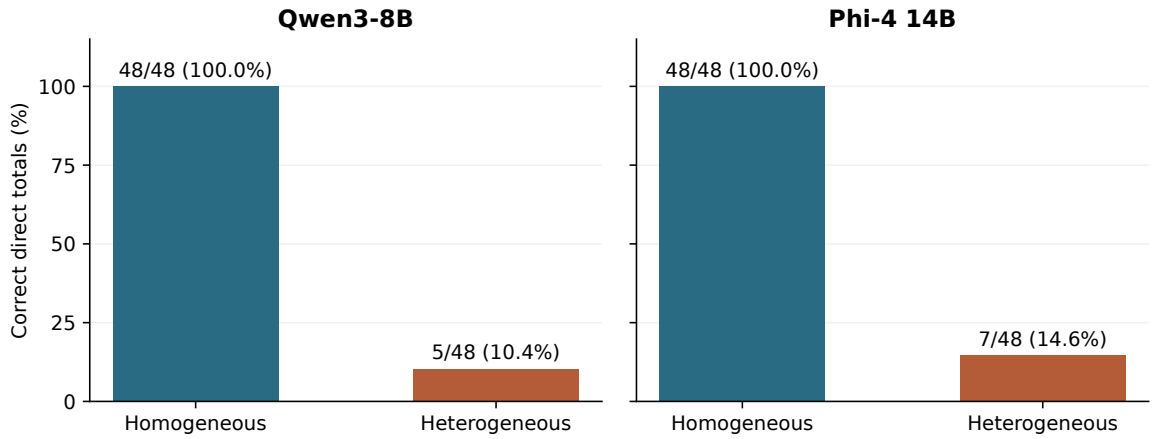


Figure 3: Correct direct totals under the original configurations. Each bar uses all 48 merged and split prompts for the indicated composition, with 24 families per bar. The subsequent Qwen recheck uses only the 24 merged controlled prompts per composition.

Both configurations passed every identical-allocation question. The varied-allocation questions yielded 5/48 correct totals for Qwen and 7/48 for Phi (Figure 3). Splitting a lot had different observed effects on the two runs. Qwen scored 4/24 on merged cases and 1/24 on split cases. Phi scored 2/24 and 5/24 respectively. This direction change limits the case for a general split-position penalty.

Field accuracy and interface compliance. In the original controlled ledger prompts, Qwen returned all requested fields correctly on 96/96 responses and reported the correct total on 55/96. The other 41 contained correct fields with a wrong total. Phi consistently named its list `lots` in place of the requested `rows`. Its strict pipeline accepted 0/96 ledgers. A separate, post-hoc sensitivity check relabelled the key without changing any values. Under that rule all 96 Phi ledgers had correct fields, and 41 had a wrong model total. Keeping the strict and relabelled scores separate makes the effect of the interface visible.

The gap in Figure 3 raised a further empirical question. Would the same Qwen checkpoint still struggle with the merged heterogeneous prompts under a configuration that allowed more computation? The 100-prompt comparison pairs old and new answers directly and retains all of the published-table cases.

6 A 100-prompt recheck of the same Qwen checkpoint

Outcome	Original	Reasoning enabled
Identical allocations, correct direct total	24/24	24/24
Varied allocations, correct direct total	4/24	24/24
Published tables, correct direct total	2/26	25/26
Published tables, correct total alongside ledger	5/26	25/26
Correct final totals, all prompts	**35/100**	**98/100**
Published tables, all reported fields correct	23/26	25/26
Calculator correct using those fields	23/26	25/26
Correct fields alongside wrong model total	18/26	0/26

Table 4. Strict, paired scores. The bottom three rows describe the same 26 ledger prompts as the fourth row. They are excluded from the 100-prompt sum.

The homogeneous-minus-heterogeneous accuracy gap was 83.3 percentage points under the original configuration. A percentile bootstrap resampling the 24 complete families gave a descriptive 95% range of 66.7 to 95.8 points. On recheck both groups reached the ceiling. Their observed gap was zero and its empirical bootstrap distribution was degenerate. Population equivalence would require evidence beyond these families. The paired outcomes in Table 5 account for the net gain of 63 correct totals: 64 previous failures became correct and one previous success failed.

Paired final-total scores	Both correct	Gained	Lost	Both failed
Controlled direct (48)	28	20	0	0
Published tables, direct (26)	2	23	0	1
Published tables, with ledger (26)	4	21	1	0

Table 5: Transitions for the same question IDs. A gain means that only the reasoning-enabled configuration was correct. Its one lost success also counts as a failure in Table 4.

Both remaining failures. The two unsuccessful responses came from the split version of the same constructed BlackRock MyMap 4/MyMap 5 account. The task asked first for a direct total and separately for a ledger. Each response consumed all 4,096 output tokens and ended with an empty final answer. The two parent-fund target percentages were 17.89% and 17.88%, so every convex average of them lies within the 0.5-point scoring tolerance. Here the measurable failure was producing the required final answer within budget. This single account provides too little variation to attribute the output-limit failures to splitting.

What changed across the matched pairs. The configurations tied on the 24 identical-allocation controls. Varying the allocations revealed a difference of 20 correct answers on 24 questions, even though the reference total in each pair was identical. The varied inputs require more arithmetic as well as sensitivity to position size. The result shows what the test can distinguish. A ledger provides additional evidence about an individual wrong total.

7 Locating an error in a published-table answer

Two recorded accounts show why the same wrong total can call for different corrections. Both use the Vanguard U.S. Equity Index Fund as the target. Every number below comes from a saved ledger response or from recomputing that response.

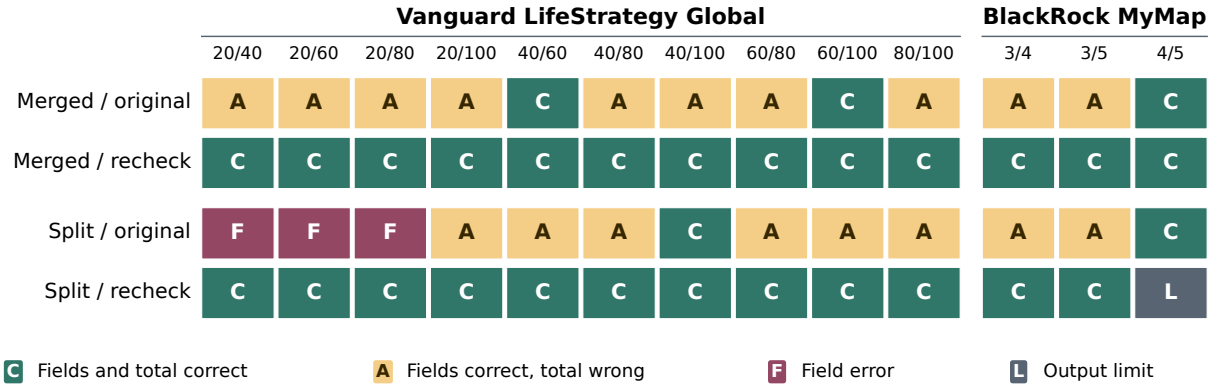


Figure 4: Strict outcomes for all 26 published-table ledger prompts under each Qwen configuration. Columns identify the pair of parent funds in a constructed account. Matching cells across runs compares the same presentation. The two direct-answer prompts per account are scored separately in Table 4.

In Figure 4, cells labelled A and F require different interventions. Recalculation can recover a total in an A cell because its reported fields match the supplied table. An F cell requires checking the assignment of fields against the source. The following two responses show the difference.

Correct fields, wrong total. A constructed account invests GBP 6,900 in LifeStrategy Global 40% Equity and GBP 3,100 in Global 80% Equity. The supplied tables give target weights of 6.8% and 28.3%. Both configurations reported those four numbers correctly. The original response reported a total of 14.6%. Its own rows imply

$$0.69(6.8) + 0.31(28.3) = 13.465\%.$$

The reasoning-enabled response reported 13.465%. Across the 26 original ledger prompts there were 18 correct ledgers with an incorrect reported total. Recomputing the reported rows recovered the correct total in all 18. The inconsistency can be checked directly from the numbers in the answer.

Wrong allocation attached to a lot. Another account splits a Global 20% position into GBP 1,860 and GBP 1,240 and invests GBP 6,900 in Global 60%. Its three target percentages should be $(0, 0, 14.8)$. The original answer reported $(14.8, 14.8, 14.8)$ and a 14.8% total. Correct arithmetic on those rows also gives 14.8%. The reference is 10.212%. The recheck reported the three source percentages and the total correctly. The original source run contained three field errors, all on split accounts involving the supplied zero-target table. Their shared inputs limit any inference about the frequency or cause of such errors.

Parser baseline. A parser tailored to the supplied table format recovered the correct totals on all 52 published-table prompts in the original evaluation. This establishes that the standardized text contains sufficient information for ordinary software to solve these cases. Evaluating extraction from raw PDFs would require a separate document-level test.

8 Interpretation and scope

Changing the product allocations and splitting a funded lot answer different questions. The first change tests sensitivity to invested amounts. The second tests whether the result survives a bookkeeping change with the same holdings. The controlled recheck compares product allocations using one presentation. The published-table prompts include both presentations, but their two capped responses concern the same account. Those observations warrant a targeted follow-up on output completion with repeated lots.

Relation to previous work. FinQA evaluates numerical reasoning over financial reports [3], while TASER addresses extraction from holdings tables [4]. Our experiment starts from supplied text and asks which portfolio inputs make a weighting error visible. The construction follows the logic of behavioural testing with controlled changes [8, 10]. Program-aided methods delegate calculation to an interpreter [5]. Here the calculator also serves as a diagnostic because it acts on the exact numbers the model reports, so a reader can check whether recomputation alone fixes the answer.

Limits. The accounts are constructed and reuse eight funds from two issuers. The source tables were transcribed and checked internally with AI assistance. We reused previously examined prompts and collected one response per configuration and prompt. The final 31 responses ran on a replacement CPU host after verification of the model and inference-program hashes. Changes to sampling, reasoning, output budget and execution occurred jointly. Estimating their individual effects would require separate interventions and repeated draws. Fresh issuers and independently checked transcriptions would test transfer beyond these documents.

Conclusion. A portfolio total is informative about weighting only to the extent that plausible weighting errors would change it. Matched heterogeneous holdings exposed a difference of 20 correct answers between two configurations that both passed every homogeneous control. On the published tables, the ledger supplied a second check. The earlier run often copied the input fields faithfully yet reported a total inconsistent with them. Together these tests specify what the observed score establishes and what should be inspected when a portfolio answer fails.

Reproducibility and disclosure. The retained research archive contains the exact prompts, paired raw answers, scoring code, source manifest and failure census. Reference totals were recomputed for all 100 prompts. AI tools assisted with design, implementation, source checks and drafting.

References

[1] bartowski. Phi-4-GGUF, 2025. URL https://huggingface.co/bartowski/phi-4-GGUF. Pinned revision and SHA-256 in the accompanying release.

[2] BlackRock. MyMap 3, MyMap 4 and MyMap 5: Monthly factsheets, 2026. URL https://www.blackrock.com/uk/literature/fact-sheet/mymap-3-class-s-inc-dist-gbp-factsheet-gb00bfbfyh34-gb-en-individual.pdf. September 2026. Class S Inc for MyMap 3 and 4; Class D Inc for MyMap 5. Top 10 Holdings on PDF page 3. Exact source URLs and document hashes in sources/source_manifest.json.

[3] Z. Chen, W. Chen, C. Smiley, S. Shah, I. Borova, D. Langdon, R. Moussa, M. Beane, T.-H. Huang, B. Routledge, and W. Y. Wang. FinQA: A dataset of numerical reasoning over financial data. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3697–3711, 2021. doi: 10.18653/v1/2021.emnlp-main.300. URL https://aclanthology.org/2021.emnlp-main.300/.

[4] N. Cho, K. Fielding, W. Watson, S. Ganesh, and M. Veloso. TASER: Table agents for schema-guided extraction and recommendation. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 5: Industry Track)*, pages 226–252. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.eacl-industry.17. URL https://aclanthology.org/2026.eacl-industry.17/.

[5] L. Gao, A. Madaan, S. Zhou, U. Alon, P. Liu, Y. Yang, J. Callan, and G. Neubig. PAL: Program-aided language models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 10764–10799, 2023. URL https://proceedings.mlr.press/v202/gao23f.html.

[6] Hugging Face Team and ggml-org. SmolLM3-3B and its GGUF conversion, 2025. URL https://huggingface.co/ggml-org/SmolLM3-3B-GGUF. Q4_K_M, pinned revision and SHA-256 in the release.

[7] Microsoft Research. Phi-4 model card, 2024. URL https://huggingface.co/microsoft/phi-4. 14B upstream model; experiment uses pinned third-party Q3_K_S quantization by bartowski.

[8] S. I. Mirzadeh, K. Alizadeh, H. Shahrokhi, O. Tuzel, S. Bengio, and M. Farajtabar. GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=AjXkRZIvjB.

[9] Qwen Team. Qwen3-8B-GGUF, 2025. URL https://huggingface.co/Qwen/Qwen3-8B-GGUF. Q4_K_M, pinned revision and SHA-256 in the release.

[10] M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.442. URL https://aclanthology.org/2020.acl-main.442/.

[11] Vanguard. LifeStrategy Global Funds: Combined quarterly report, 2026. URL https://www.vanguard.co.uk/content/dam/intl/europe/documents/en/lifestrategy-global-fund-combined-quarterly-report.pdf. 30 September 2026, underlying-funds tables on PDF pages 6, 10, 14, 18, and 22.